

Michaela A. Polley

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EDUCATION

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| 2029 (exp.) | DARTMOUTH COLLEGE, Hanover, New Hampshire
Ph.D. Mathematics (Advisor Sergi Elizalde) |
| 2023 | CARLETON COLLEGE, Northfield, Minnesota
B.A. Mathematics, Chemistry (Advisors Eric S. Egge, Daniela Kohen) |

APPOINTMENTS

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| 2023 - 24 | ALPEN-ADRIA-UNIVERSITÄT, Klagenfurt, Austria
Visiting Scholar |
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GRANTS

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| 2023 - 24 | <i>Fulbright-Austrian Marshall Plan Foundation Award</i> , Alpen-Adria-Universität |
| 2020 | <i>Summer Science Fellow</i> , Carleton College |

AWARDS AND FELLOWSHIPS

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| 2023 | <i>Steven P. Galovich Prize in Mathematics</i> , Carleton College |
| 2021 - 22 | <i>Dean's List</i> , Carleton College |
| 2019 | <i>National Merit Scholar</i> , St. Paul Academy and Summit School |

PUBLISHED ARTICLES

- 1 *Mechanical Recycling of 3D-Printed Thermosets for Reuse in Vat Photopolymerization.* (with Erin M. Maines, Greg Haugstad, Brenda Zhao, Theresa M. Reineke, and Christopher J. Ellison), ACS Applied Polymer Materials, vol. 6, num. 8 (2024), 4625–4633. <https://doi.org/10.1021/acsapm.4c00184>
- 2 *The 334-Triangle Graph of $SL_3(\mathbb{Z})$.* (with Eric Egge), Involve: A Journal of Mathematics, vol. 15, num. 3 (2022), 537-546. <https://doi.org/10.2140/involve.2022.15.537>
- 3 *Patterns in rectangulations. Part I: T-like patterns, inversion sequence classes $I(010,101,120,201)$ and $I(011,201)$, and rushed Dyck paths.* (with Andrei Asinowski), Discrete Mathematics & Theoretical Computer Science, vol. 27, num. 1, Permutation Patterns 2024 (2025), <https://doi.org/10.46298/dmtcs.15118>

INVITED CONFERENCE TALKS

- 2026 Mar *Characterization of Avoidance of Patterns in Rectangulations by Permutation Mesh Patterns*
AMS Spring Sectional Meeting Special Session on Permutation Patterns; Boston, MA
- 2025 Apr *Patterns in Rectangulations: T-avoiding rectangulations, Catalan structures, inversion sequences, and Dyck paths*
AMS Spring Sectional Meeting Special Session on Geometric and Algebraic Combinatorics; Hartford, CT

CONTRIBUTED CONFERENCE TALKS

- 2025 Jul *Characterization of Avoidance of One-Sided 2- and 3- Segment Patterns in Rectangulations by Mesh Patterns*
Permutation Patterns 2025; St Andrews, Scotland
- Mar *Patterns in Rectangulations: T-avoiding rectangulations, Catalan structures, inversion sequences, and Dyck paths*
50th Annual New York State Regional Graduate Mathematics Conference; Syracuse, NY
- 2024 Jun *T-avoiding rectangulations, inversion sequences, Dyck paths*
Permutation Patterns 2024; Moscow, ID
- 2022 Jan *The 334-Triangle Graph of $SL(3, \mathbb{Z})$*
Nebraska Conference for Undergraduate Women in Mathematics; virtual
- 2020 Oct *Enumeration of Minimal 2-Cuts on Surfaces* (with Soren DeHaan)
Northfield Undergraduate Mathematics Symposium; virtual
- Aug *Enumeration of Minimal 2-Cuts on Surfaces* (with Soren DeHaan)
Young Mathematicians Conference; virtual

RESEARCH SEMINAR TALKS

- 2026 Jan *Patterns in Rectangulations*
Dartmouth Graduate Student Seminar; Hanover, NH
- 2025 Dec *Patterns in Rectangulations*
Doctoral Seminar; Klagenfurt, Austria
- Apr *Patterns in Rectangulations: T-like patterns, Inversion Sequences, and Dyck Paths*
Dartmouth Combinatorics Seminar; Hanover, NH

CONFERENCE POSTER SESSIONS

- 2022 Oct *Reverse Engineering Mass Spectrometry*
Carleton Undergraduate Research and Internship Symposium; Northfield, MN
- 2021 Oct *Recycling Photopolymerizable 3D Printing Resin*
Carleton Undergraduate Research and Internship Symposium; Northfield, MN
- Aug *Recycling Photopolymerizable 3D Printing Resin*
Summer Undergraduate Research Expo; Minneapolis, MN
- 2020 Oct *Enumeration of Minimal 2-Cuts on Surfaces*
Carleton Undergraduate Research and Internship Symposium; virtual

UNDERGRADUATE RESEARCH TALKS

2025 Sep *Patterns in Rectangulations*
Women in Mathematics in New England; Northampton, MA

WORKSHOPS/RESEARCH VISITS

2025 Jul *Permutation Patterns 2025 Pre-Conference Workshop*, St Andrews, Scotland
2024 Mar *Combinatorics, Algorithms, and Geometry*, Dresden, Germany

CONFERENCE ORGANIZING

2027 Jul *Permutation Patterns 2027*, Klagenfurt, Austria

GRADUATE READING COURSES/LEARNING SEMINARS DIRECTED

2026 Win Dartmouth Combinatorics Learning and Working Seminars (co-director)
Topic: Coxeter Groups and the Bruhat Order

TEACHING

2026 Win *Math 13: Calculus of Vector-valued Functions* (Teaching Assistant), Dartmouth
2025 Win *Math 8: Calculus of Fun. of One and Several Variables* (Teaching Assistant), Dartmouth
2024 Fall *Math 8: Calculus of Fun. of One and Several Variables* (Teaching Assistant), Dartmouth
Sum *Math 311.919: Linear Algebra for Engineers* (Tutor), Alpen-Adria-Universität
Sum *Math 311.214: Analysis II for Engineers* (Tutor), Alpen-Adria-Universität
2023 Win *Math 311.109: Analysis I for Engineers* (Tutor), Alpen-Adria-Universität
2022 Fall *Math 101: Calculus with Problem Solving* (Teaching Assistant), Carleton
Spr *Math 241: Ordinary Differential Equations* (Teaching Assistant/Prefect), Carleton
Win *Math 101: Calculus with Problem Solving* (Teaching Assistant), Carleton

TEACHING WORKSHOPS ATTENDED

2025 Sep *STEM Boldly*
Dartmouth NEXT
Sep *Teaching Students with ADHD*
Accessible Dartmouth Initiative
Sep *Teaching Autistic Students*
Accessible Dartmouth Initiative
Jul *Learning Community for Future Faculty: Supporting Students in Difficult Times*
Dartmouth Center for the Advancement of Learning
Jun *Learning Community for Future Faculty: Constructive uses of AI*
Dartmouth Center for the Advancement of Learning
Feb *Improving Accessibility*
OURFA²M²
Feb *The Importance of Advocacy*
OURFA²M²
2024 Dec *Teaching with GenAI Faculty Showcase*
Dartmouth Center for the Advancement of Learning

Sep *Generative Artificial Intelligence and Mathematics*
Dartmouth Center for the Advancement of Learning

ACADEMIC OUTREACH

2022 - 23 Senior Fellow for Data, Joyce Ivy Foundation

ACADEMIC SERVICE

2025 - Department Tea organizer, Dartmouth College Mathematics Department
2025 - Combinatorics group meetings organizer, Dartmouth College Mathematics Department
2024 - Graduate Program Committee Liaison, Dartmouth College Mathematics Department
2022 - 2023 Student Departmental Advisor, Carleton College Mathematics Department
2021 - 2023 Board Member for Gender Minorities in Mathematics and Statistics, Carleton College
Mathematics Department

ACADEMIC ORGANIZATION MEMBERSHIPS

Sigma Xi (Associate Member), Association for Women in Mathematics, American Mathematical Society

EDITORIAL WORK

Referee for Discrete Mathematics and Theoretical Computer Science (DMTCS)